

Galaxy for virologist training. Exercise 4: Nanopore Assembly 101

Title	Galaxy
Training dataset:	Nanopore MinION Sequencing of a Monkey Pox Virus (MPXV) from Spain 2022 outbreak. Data is publicly available at SRA with ID ERR10297654 . Paper .
Questions:	• How different are Nanopore reads assembled from Illumina reads?
Objectives:	• Understand the concept of assembly. • Learn how to interpret assembly quality control metrics.
Estimated time:	40 min

1. Description

Nanopore technology is a third generation sequencing technique which enables the obtaining of longer sequences, but with reduced sequence quality. Different technologies have different formats, qualities, and specific known biases which make the analysis different among them.

In this tutorial, we will see an example of how to assemble long reads from a Nanopore sequencing run.

2. Upload data to Galaxy

Training dataset

- SRA ID: [ERR10297654](#).

Create new history

- Click on the **+** icon at the top of the history panel and create a new history with the name **Nanopore assembly 101 tutorial** as explained [here](#).

Upload data

1. Look for **SRA** in the tool search bar and select **Faster Download and Extract Reads in FASTQ format from NCBI SRA**.
2. Accession = [ERR10297654](#).
3. Run Tool.

Galaxy Europe

Tools

SRA

Upload Data

Show Sections

SRA server

Download and Generate Pileup Format from NCBI SRA

Download and Extract Reads in BAM format from NCBI SRA

Faster Download and Extract Reads in FASTQ format from NCBI SRA

pysradb search sequence metadata from SRA/ENA

EBI SRA ENA SRA

LowMemPeakPickerHiResRandomAccess Finds mass spectrometric peaks in profile mass spectra.

Make.sra creates the necessary files for a NCBI submission

Faster Download and Extract Reads in FASTQ format from NCBI SRA (Galaxy Version 2.11.0+galaxy1)

select input type

SRR accession

Accession

ERR10297654

Must start with SRR, DRR or ERR, e.g. SRR925743, ERR343809

Advanced Options

Email notification

No

Send an email notification when the job completes.

✓ Execute

What it does?

This tool extracts data (in fastq format) from the Short Read Archive (SRA) at the National Center for Biotechnology Information (NCBI). It is based on the [fasterq-dump](#) utility of the SRA Toolkit.

How to use it?

Load reference file from NCBI

1. Search **NCBI** using the search toolbox and select **NCBI Accession Download Download sequences from GenBank/RefSeq by accession through the NCBI ENTREZ API**.
2. Select source for IDs > **Direct entry**.
3. ID List = **NC_063383.1**
4. Run Tool.

Galaxy Europe

Herramientas

ncbi

Cargar Datos

Show Sections

Krona pie chart from taxonomic profile

Kraken taxonomic report view report of classification for multiple samples

Unipept retrieve taxonomy for peptides

MaxQuant (using mqpar.xml)

Krona pie chart from taxonomic profile

NCBI ESearch search NCBI Databases by text query

NCBI EPost post UIDs to NCBI History Server

NCBI EFetch fetch records from NCBI

NCBI Accession Download Download sequences from GenBank/RefSeq by accession through the NCBI ENTREZ API

NCBI EInfo fetch NCBI database metadata

NCBI ECIIMatch search NCBI for citations in PubMed

ETE tree DB generator generate the

NCBI Accession Download Download sequences from GenBank/RefSeq by accession through the NCBI ENTREZ API (Galaxy Version 0.2.7+galaxy0)

Select source for IDs

Direct Entry

ID List

NC_063383.1

Newline/Comma separated list of IDs

Molecule Type

Nucleotide

File Format

FASTA

How to handle download failures

Abort with error on first failure

Add accession to failed list and continue

Email notification

No

Send an email notification when the job completes.

✓ Execute

History

buscar conjuntos de datos

nanopore assembly 101 tutorial

361 MB

4 : fasterq-dump log

3 : Other data (fasterq-dump)

a list with 0 datasets

2 : Single-end data (fasterq-dump)

a list with 1 fastqsanger.gz dataset

1 : Pair-end data (fasterq-dump)

a list with 0 pairs

Unhide data

Using SRA and NCBI API downloads data as hidden so we are going to unhide this data as follows:

1. Click on the **strikethrough eye** (Show hidden).
2. Select the strikethrough for ERR10297654 and NC_063383.1 data.
3. Then select the location icon (show active).

History

buscar conjuntos de datos

nanopore assembly 101 tutorial

361 MB

6

2

7 : NCBI Accession Download on : Log

6 : NCBI Accession Download on : Downloaded Files

a list with 1 fasta dataset

4 : fasterq-dump log

3 : Other data (fasterq-dump)

a list with 0 datasets

2 : Single-end data (fasterq-dump)

a list with 1 fastqsanger.gz dataset

1 : Pair-end data (fasterq-dump)

a list with 0 pairs

History

visible:false

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4

6

2

8 : NC_063383.1

5 : ERR10297654

2

3

Assemble reads with Flye

1. Search for the **Flye** assembler using the search toolbox and select **Flye de novo assembler for single molecule sequencing reads**.
2. Input reads: **ERR10297654**.
3. Click on Run Tool and wait.

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Flujo de Trabajo Visualizar Datos Compartidos Ayuda Usuario

Herramientas

Flye

Cargar Datos

Show Sections

Flye de novo assembler for single molecule sequencing reads

FLUJOS DE TRABAJO

Todos los flujos de trabajo

Flye de novo assembler for single molecule sequencing reads (Galaxy Version 2.9.1+galaxy0)

Input reads

8: NC_063383.1

5: ERR10297654

Mode

Nanopore raw (--nano-raw)

Number of polishing iterations

1

Polishing is performed as the final assembly stage. By default, Flye runs one polishing iteration. Additional iterations might correct a small number of extra errors (due to improvements on how reads may align to the corrected assembly). If the parameter is set to 0, the polishing is not performed (--iterations)

Minimum overlap between reads

4 Execute

Assembly quality control with QUAST

1. Search QUAST in the search tool box.
2. **Assembly mode? > Individual assembly.**

- Contigs/scaffolds file > **Flye results**.
- Use a reference genome: **Yes**. Select the **NC_063383.1 FASTA file** previously loaded.
- Run Tool.

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Flujo de Trabajo Visualizar Datos Compartidos Ayuda Usuario

Herramientas

quast 1

Cargar Datos

Show Sections

Quast Genome assembly Quality 2

rnaQUAST A Quality Assessment Tool for De Novo Transcriptome Assemblies

FLUJOS DE TRABAJO

Todos los flujos de trabajo

Quast Genome assembly Quality (Galaxy Version 5.2.0+galaxy0)

Use customized names for the input files?

No, use dataset names

They will be used in reports, plots and logs

Contigs/scaffolds file

11: Flye on data 5: consensus
8: NC_063383.1

3

Reads options

Disabled

Currently, the supported read types are Illumina unpaired, paired-end and mate-pair reads, PacBio SMRT, and Oxford Nanopore long reads.

Type of assembly

Genome

Use a reference genome?

Yes

Many metrics can't be evaluated without a reference. If this is omitted, QUAST will only report the metrics that can be evaluated without a reference.

Reference genome

11: Flye on data 5: consensus
8: NC_063383.1

5

6 Execute

- Click on the eye icon from the **QUAST HTML report**.

- How much of or reference genome have we reconstructed?
- How many contigs do we have greater than 1000 pb?
- How long is the largest contig in the assembly?
- Which is the N50?

- Open the **Icarus** viewer in the QUAST report.

QUAST
Quality Assessment Tool for Genome Assemblies by CAB

17 November 2021, Wednesday, 18:07:45

[View in Icarus contig browser](#)

All statistics are based on contigs of size ≥ 500 bp, unless otherwise noted (e.g., "# cor bp)" and "Total length (≥ 0 bp)" include all contigs).

Aligned to "dataset_76f460df_9dce_4919_9108_be70c4d29af9" | 11 029 bp | 1 fragment G+C

Genome statistics **SPAdes_on_data_3_and_data_2_...**

Genome fraction (%) 83.070

- Which contig align against our reference genome?

This training history is available at: <https://usegalaxy.eu/u/s.varona/h/nanopore-assembly-101-tutorial>

NOTE: Nanopore data is known to have a higher error rate than short sequencing reads. This is why assembly post-processing is strongly recommended, usually using combined sequencing approximation with both Nanopore and Illumina reads.